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Stacking Ensemble Technique to Predict Cervical Cancer Using Hyperparameter Tuning and Feature Selection

Md. Tofael Ahmed Bhuiyan
Dept. of Computer Science and Engineering
Southeast University
Dhaka, Bangladesh
tofael1104@gmail.com

Khandaker Mohammad Mohi Uddin*
Dept. of Computer Science and Engineering
Southeast University
Dhaka, Bangladesh
jilanicsejnu@gmail.com

Md. Rakibul Islam
Dept. of Computer Science and Engineering
Southeast University
Dhaka, Bangladesh
Rakibul.3200@gmail.com

Md. Habibullah Belali
Dept. of Computer Science and Engineering
Southeast University
Dhaka, Bangladesh
habibullah.belali@seu.edu.bd

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***Abstract*— Cervical cancer is the primary cause of premature mortality for women in underdeveloped countries, where it causes approximately 85% of cases. Better outcomes from treatment and preventative measures depend on early risk factor assessment. A thorough prediction model based on preliminary testing data and individual medical records is presented in this study. The risks were analyzed using algorithms such as Random Forest (RF), Decision Tree (DT), and AdaBoost. SelectKBest, XGBoost, and Chi-Square are examples of feature selection approaches that improved feature extraction, improving performance and interpretation. The algorithms for machine learning that were used were LR, LGBM, SVM, KNN, and Naive Bayes. To attain the highest accuracy, grid search and cross-validation were used for hyper parameter optimization. Stacking ensembles combined RF, DT, and AdB, achieving 99.443% accuracy. Ensemble approaches, particularly when resolving decreases in features and disparity in class, helped to make clinical decisions and accurately predicted the risk of cervical cancer.**

Keywords—Cervical Cancer, Machine Learning, Hyperparameter Optimization, Feature Selection, Stacking Ensemble Approach.

I. INTRODUCTION

The most prevalent gynecological cancer, cervical cancer is an important health issue that is mostly brought on by carcinogenic variants of the human papillomavirus (HPV). About 485,000 individuals are identified annually, making it the third most common cancer worldwide. It also contributes to more than 236,000 deaths every year, making it the fourth most common cause of cancer-related mortality [1,2].It was predicted that 604,000 more instances of cervical cancer will be diagnosed by 2020, solidifying its position as the third most common disease in women globally. Sadly, it was predicted that 90% of the anticipated 34,200 fatalities will occur in middle- to low-income nations. HIV accounts for 5% of cases worldwide, and women who have the virus are six times more vulnerable [3,4].

The main contributory factor of cervical cancer is long-term exposure to high-risk human papillomavirus (HPV) strains. There are more than 200 HPV variations, both low-risk and high-risk. The latter are associated with a number of malignancies, such as those of the cervical, vulvar, vaginal, penile, anal, and oropharyngeal regions. Health problems are usually caused by low-risk strains. Although skin-to-skin contact is the primary method of HPV transmission, other forms of intimate contact can also spread the virus. Because of vaccinations that target high-risk HPV strains, the incidence of malignancies linked to HPV has drastically decreased. Nonetheless, chronic oncogenic HPV infection continues to be a major cause, underscoring the necessity of prevention in addition to public health campaigns and quit smoking efforts.

Although the significance of data preparation is occasionally disregarded, recent research on cervical cancer has placed an increasing emphasis on the creation of more precise forecasting models. To resolve data discrepancies and enhance model dependability, effective planning is essential, particularly for outlier detection. The precision of making choices can be improved by using more robust preprocessing methods for outlier identification. One of the best methods for predictive modelling, particularly in cancer research, has been machine learning (ML). Cervical cancer diagnosis and prediction have been accomplished by a number of studies, demonstrating the revolutionary abilities of machine learning techniques in early identification and improving outcomes for patients [5, 6, 7].

The current research has benefited from the following major contributions. It is now easy, reliable, and efficient to forecast cervical cancer using a machine learning-based method. Also, the following are the primary goals of our methodology:

- A realistic cervical cancer prediction is proposed by this study using 14 significant clinical and genetic variables.
- In order to identify significant and related biomarkers, 9 machine learning techniques were applied.
- Feature scaling, hyperparameter manipulation, and the voting classifier are used to maximize model performance.
- The Random Forest, Decision Tree, and AdaBoost models performed better than the others, with the greatest prediction accuracy of 99.44%.
- Other noteworthy model configurations that demonstrate the importance of ensemble approaches in improving

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diagnostic reliability are Naive Bayes, KNN, CatBoost, Logistic Regression, and the SVM and LGBM algorithms.

II. Related Works

The next section explores a number of newly developed methods for cervical cancer prediction, classifying them as machine learning-based techniques.

Mohiuddin et al. [8] proposed an advanced ensemble learning system for the diagnosis of cervical cancer, including methods such as ROS, XGBoost, PCA, and a hard voting mechanism that combined RF and MLP. Their model was able to achieve the highest accuracy of 99.19% by analysing 858 patient records that included 36 features, demonstrating that ensemble techniques work rather effectively in handling class imbalance with regard to improved diagnostic outcomes. Tanimu et al. [9] proposed a decision tree-based model for risk analysis in cervical cancer, giving considerable emphasis on feature selection-as done with RFE and LASSO. The resultant approach formulated by them substantially improved both predictive accuracy and the stability of the model, showing how judicious feature engineering remains an essential activity in clinical diagnostics.

Abdoh et al. [10] proposed the Random Forest classifier combined with SMOTE to improve the diagnosis of cervical cancer. For dimensionality reduction, RFE and PCA were applied. From their results, the proposed SMOTE-RF outperformed SVM, further underlining the importance of resampling when handling complex medical datasets. Ijaz et al. [11] proposed a novel hybrid approach that combined DBSCAN, SMOTE, RF, and Tomek linkages for the diagnosis of cervical abnormalities in 858 ASCUS patients. Its integrated architecture succeeded in noise reduction and quality improvement of prediction with an outstanding accuracy of 97.72%. This exciting advantage of hybrid machine learning approaches increases model resilience and diagnostic accuracy.

Ali et al. [12] used a Random Forest classifier to achieve 98.33% accuracy on the cervical cancer dataset they had downloaded from Kaggle. This outcome shows how well Random Forest works with complex clinical data. Alsamaiz et al. [13] used a vote algorithm on a clinical dataset in another investigation, and the accuracy was 97.44%. This suggests that ensemble methods offer a reliable way to combine estimates from several classifiers to get better results. Their results provide credence to this conclusion. Abdulaziz et al. developed a stacked ensemble framework [14]. They combined a lot of fundamental classifiers. In this manner, their accuracy was 99.44%.

III. Materials and Methods

A. Material and Method

In order to convert the dataset into numerical form, we meticulously prepared it by addressing missing data and encoding categorical values. Resampling algorithms maintained equitable representation, and key characteristics were chosen to draw attention to significant parts. The dataset was divided 80/20 between testing and training. Machine learning algorithms were constructed using the training set and assessed using the test set. The matrix of confusion in the final ensembles with a voting classifier was used for evaluating performance. Results were compared to determine which model performed the best. Intentionally linguistic errors were eliminated to prevent ethical dilemmas, guaranteeing impartial and accurate forecasts in the cervical cancer investigation.

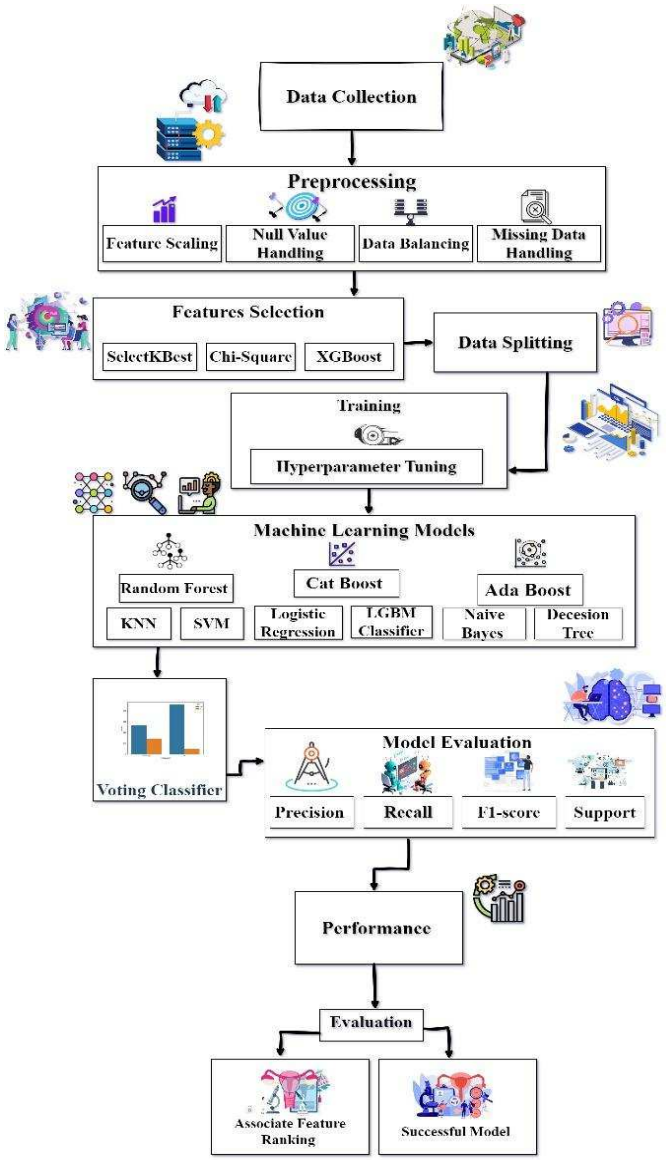


Fig.1: Suggested cervical cancer system model.

B. Datasets

The study's associated risk factors dataset is publicly available via the UCI Machine Learning Repository. It was obtained from Caracas University Hospital in Caracas, Venezuela [15]. It contains the medical histories, lifestyle choices, and demographic information of 858 individuals, each of whom has 36 traits. In order to maintain accurate data, missing values must be handled properly. Important warning signs for cervical cancer are highlighted by the dataset's attributes. The factors to be examined include the results of examinations that are crucial for identifying the presence of

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cervical cancer, including cytology, biopsy, Schiller, Himmelman, and others.

C. Dataset preprocessing

Before combining datasets with machine learning models, data preparation is essential to guarantee accuracy. By reducing dimensionality, Principal Component Analysis (PCA) increases efficiency. In accordance with Equations (1) and (2), the mode for categories characteristics and the mean for numerical factors are used to fill in the missing values. This procedure preserved data integrity by reducing the number of columns from 858 to 737.

$$\boldsymbol{MeanX} = \frac{\mathbf{1}}{\boldsymbol{n}}\left(\sum_{\boldsymbol{i=1}}^{\boldsymbol{n}} \boldsymbol{x_i}\right), \quad \textbf{(1)}$$

where,

X_i= *I-th variable*

I = *I-th value of variable X*

N = *Number of variables in the dataset (*n >0)

$$\boldsymbol{Mode\ Z} = \boldsymbol{L} + \left(\frac{\boldsymbol{f_m}+\boldsymbol{f_{m-1}}}{(\boldsymbol{f_m}-\boldsymbol{f_{m-1}})+(\boldsymbol{f_m}-\boldsymbol{f_{m-1}})}\right)\boldsymbol{C} \quad \textbf{(2)}$$

While $\boldsymbol{m}$ indicates the mode class itself, L represents the mode class's bottom border.$\boldsymbol{f_m-1}$ indicates the rate of the previous class, whereas $\boldsymbol{f_m}$ represents the frequencies of the mode class.$\boldsymbol{f_m+1}$ represents the frequency of the next class, and $\boldsymbol{C}$ represents the mode class's length. Furthermore, features that had missing values for more than 60% of their beliefs, including STDs_Time_since_last_diagnosis and STDs_Time_since_first_diagnosis (787/858), were removed from the dataset.

D. Feature Scaling

Predictive modelling requires feature scaling, particularly for SVM and KNN. To guarantee that each feature contributed proportionately, we employed sophisticated normalization and standardization. By evaluating categorical elements, chi-square feature selection improved predictive power. Preprocessing using SelectKBest and XGBoost increased the models' accuracy and processing capacity, guaranteeing outstanding outcomes.

E. Data Splitting

In a successful attempt to ensure that models would be effective when generalized, the data was separated into an 80-20 ratio to constitute the training set and test set.

F. Hyper parameter Tuning

Hyper parameter tuning seeks to improve the performance of a model by adjusting parameters with a structured approach. Both studies used Grid Search and Random Search to investigate parameters. In the cervical cancer study, the ensemble models, Random Forest and XGBoost, were further improved by adjusting the additional parameters such as estimators, depth, learning rate in order to gain better accuracy, sensitivity, and specificity. In the case of SVM, KNN, and CAT Boost, cross-validation was added in our classification study in order to refine their hyper parameters. The Chi-Square feature selection proves unique diligence rather methodical ambiguity. These methods produce machine learning models that are reliable, robust, and scalable and that can withstand hard real-life challenges effectively.

G. Correlation Coefficient

The relationship or connection between two variables can be measured using a correlation function (3). In particular, correlation measures the relationship between changes in one variable and changes in another. The intensity and direction of this association are indicated by the correlation coefficient, which has a range of -1 to 1. Strong negative correlations are suggested by values near -1, lesser relationships are implied by values around 0, while the strong positive correlations are indicated by values close to 1. The relationship between variables K and M is described as follows in [16]:

$$\boldsymbol{\rho_{K,M}} = \frac{\boldsymbol{cov(K,M)}}{\boldsymbol{\sigma K \sigma M}} \quad \textbf{(3)}$$

H. Label Encoding

Label encoding is a frequently employed technique for handling categorical data. With this method, every classification is given a unique integer value determined by the labels' ordinal order. The following formulae (4) and (5) are used to determine the encoded values.

$$\boldsymbol{S} = \frac{\mathbf{1}}{\mathbf{1} + \boldsymbol{exp}(-\frac{\boldsymbol{n-mdl}}{\boldsymbol{a}})} \quad \textbf{(4)}$$

$$\hat{\boldsymbol{x}}^{\boldsymbol{k}} = \boldsymbol{prior} * (\mathbf{1} - \boldsymbol{s}) + \boldsymbol{s} * \frac{\boldsymbol{n}^{+}}{\boldsymbol{n}} \quad \textbf{(5)}$$

The optimization parameter α is used in this context to assess the regularization performance, while $\boldsymbol{md}$l stands for the smallest quantity of data elements needed at a leaf node. It is reasonable to suppose that both $\boldsymbol{mdl}$ and a have values in the range of 1 and 100. The old values are replaced with new values unique to the training dataset, as explained in [17].

I. Missing Data Handling

Some instances of missing information are visible in the clinical dataset, which includes 858 patients and 36 highlights. To fill up these gaps, this study employed the cruel ascription technique, which enables analysts to determine the average of available values and replace the missing parts with this harsh. This strategy ensures that the dataset is complete while reducing the bias caused by missing data, as shown in Equation (6).

$$\boldsymbol{Mean} = \frac{\boldsymbol{Sum\ of\ all\ data\ points}}{\boldsymbol{Number\ of\ data\ points}} \quad \textbf{(6)}$$

Four objectives are examined in this study: cytology, biopsy, Handelman, and Schiller tests. After applying iodine to the cervix, the Schiller test visually assesses the malignancy. However, cytology is mostly employed since the Schiller test lacks sensitivity and specificity. Cytology tests for infections of the urinary tract, tumors diseases, and cervical cancer. The Handelman evaluation includes a comprehensive assessment of the cervix, vulva, and vagina.

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J. Feature Selection

Feature selection is a crucial part of the process that involves identifying required attributes to reduce the dimensionality of input characteristics. Decreased data complexity and improved model performance and accuracy can be achieved by reducing the number of input parameters [18, 19]. In this study we applied the SelectKBest method to analyze the clinical ovarian cancer data, then try to select relevant features. The scoring function—Chi-Square—is used to score a certain feature, then XGBoost is used to rank the feature in order to have it well assessed. These methods are assembled together to provide a fair evaluation of the significance of a feature on the model performance.

K. Standard Scaler

The Standard Scaler is a technique that normalizes dataset properties by converting them to have a zero mean and one standard deviation. Data that has been standardized is suitable for modeling and ensures that no characteristic will have a significant effect on the algorithm due to scale differences. In linear models, Standard Scaler's feature standardization makes it easier to understand the coefficients or weights assigned to each feature. In equation (7), the Standard Scaler is expressed mathematically.

$$X_{new} = \frac{X_i - X_{mean}}{X_{sd}} \quad (7)$$

L. Machine Learning Algorithm

Classification is an essential component of machine learning methods used worldwide for illness diagnosis [20]. The following section provides a summary of how several classifiers work: Support Vector Machines (SVM), Random Forest (RF), Decision Trees (DT), Naive Bayes (NB), CAT Boost, XGBoost (XGB), LGBM, Logistic Regression (LR), AdaBoost, and Logistic Regression.

IV. Result and Discussion

This study evaluates cervical cancer prediction algorithms using metrics such as F1 score, accuracy, recall, and precision. In order to enhance model performance and identify optimal parameters, cross-validation was employed. Several methods were used to change variables and lessen overfitting. A confusion matrix was employed to conduct a complete performance study of both linear and multiclass activities. To predict and categorize cervical cancer, a machine learning classifier was created.

A. System Requirements

Data processing and model construction were carried out in an advanced optimized configuration that ensured the highest accuracy and efficiency for challenging machine learning techniques. To prevent bias from entering the analyses or leading to data breaches, the data were meticulously prepared, pre-processed, and shuffled. Additionally, the core system was fantastic. With an Intel® Xeon® CPU, 12.68GB of RAM, and a Tesla K80 GPU, this two-core computer managed demanding tasks with ease. For testing and any compatibility added to the model's efficiency, an independent Intel® Core i3 with 8.00GB was helpful; it was run on the different hardware configurations. This technique was also developed using libraries that are still in use today, including NumPy, Scikit-Learn, and Panda.

B. Confusion matrix

To evaluate the results, recall, F-measure, accuracy, and precision—all based on the classification report and confusion matrix—were applied. The outcome is greatly influenced by the precision of the model's training procedure. Equations (8) through (11) describe how indicators for assessment were computed using performance measurements like true positive (TP), false positive (FP), true negative (TN), and false negative (FN) numbers. These measurements provide a thorough assessment of the model's ability to detect cervical cancer properly.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (8)$$

$$Precision = \frac{TP}{TP + FP} \quad (9)$$

$$Recall = \frac{TP}{TP + FN} \quad (10)$$

$$F1 = 2 * \frac{Precision.Recall}{Precision + Recall} \quad (11)$$

The most crucial criteria in medical diagnostic studies are precision and recall. Because it combines memory and accuracy into a single, complete score, the F-measure is useful for assessment. It is important to emphasis recollection since false negatives (FNs) might be more harmful than false positives (FPs). While FNs can have significant effects, FPs are frequently disregarded. "Fig.2" displays the model's confusion matrix.

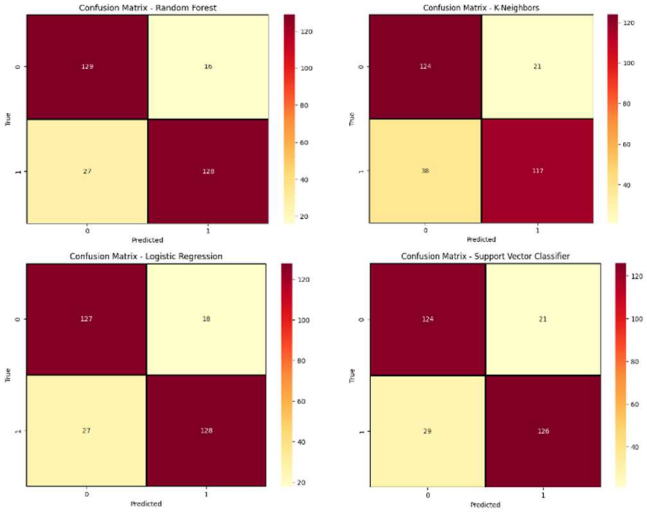


Fig.2: Confusion Matrix of the model (a) Random Forest (b) K-Neighbors c) Logistic Regression (d) Decision Tree

C. Stacking Ensemble

Stacking ensembles are crucial for increasing machine learning models' accuracy. Nine Classifiers were merged into a

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stacking strategy that successfully recognized cervical cancer in this study with a 97.83% success rate. A remarkable 97.83% accuracy rate in cervical cancer diagnosis was attained by the stacking method that combined those algorithms.

The efficiency indicators of several classifier combinations for identifying cervical cancer are described in detail in Table 1. When combined with superior precision, recall, and F1 scores, the RF + DT + AdaBoost ensemble achieves a remarkable accuracy of 0.99443. Even though the combination in question is better, RF combined with SVM or LR also produces excellent outcomes. Metrics like recall, accuracy, precision, and F1 scores vary among classification groups; RF + SVM and RF + AdaBoost outperform them with accuracy and precision of 99.47% and 98.70%, respectively. With an impressive 0.9710 precision and 99.10% accuracy, the RF + LR + SVM + DT ensemble scores well on all measures. For a precise diagnosis, ensembles—particularly RF, DT, and AdaBoost—are essential.

D. Computational Complexity and Scalability

The primary models in the suggested stacking ensemble are the source of its computational difficulty. AdaBoost (AdB), Random Forest (RF), and Decision Tree (DT) have complexities of *O(n.M.T), O(M.n.logn)*, and *O(n.logn*), respectively, with AdB being the most computationally intensive. While shared training was made possible by the parallelism characteristics of RF and AdB, scalability was assessed utilizing multi-core CPUs and GPUs. In order to guarantee effectiveness and flexibility across computer systems, memory complexity—which is impacted by tree depth and estimator count—was minimized by hyper parameter customization.

Table-1: Performance Metrics for Classifier Combinations of Ensemble

Combination	Accuracy	Precision	Recall	F1 Score
RF + SVM	0.994161	0.9864	0.97826	0.9711
RF + LR	0.994161	0.96414	0.978261	0.9809
RF + DT	0.988261	0.976994	0.97826	0.9478
RF+ AdaBoost	0.97413	0.9870	0.9783	0.9675
DT + AdaBoost	0.9939	0.9870	0.9783	0.9675
LR + SVM	0.988261	0.956994	0.97826	0.96751
LR + DT	0.993461	0.9493	0.97826	0.9823
SVM + DT	0.978261	0.95699	0.97821	0.96751
RF + DT + AdaBoost	0.99443	0.9870	0.9783	0.9675
RF + LR + SVM	0.992761	0.95699	0.97826	0.9456
RF + SVM + DT	0.993861	0.95699	0.97826	0.96751
LR + SVM + DT	0.988261	0.95699	0.97826	0.9543
RF + LR + SVM + DT	0.993961	0.95699	0.97826	0.9492
RF + LR + SVM + DT + LGBM + CAT + KNN + NB	0.991014	0.95683	0.97101	0.9414

Table 2 shows an assessment of each model's accuracy in order to minimize overfitting. Additionally, it describes each algorithm's F1-score, accuracy, recall, and precision. The ROC curve for various machine learning classifiers for testing data is shown in "Fig.3."

TABLE-2: PERFORMANCE EVALUATION OF SEVERAL ML MODELS

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	99.44%	0.99	0.99	0.99
AdaBoost	99.39%	0.97	0.97	0.97
Logistic Regression	98.81%	0.92	0.91	0.91
Decision Tree	99.41%	0.97	0.97	0.97
LGBM	99.39%	0.97	0.97	0.97
Naïve Bayes	98.67%	0.99	0.98	0.99
SVM	98.78%	099	0.99	0.99
KNN	98.83%	0.99	1.00	1.00
Cat Boost	98.78%	0.99	0.98	0.98

The results of the cervical cancer test dataset are shown in "Fig.3.", which also highlights the attributes that the model chose to have the lowest squared error. SGD learning was used to efficiently impute all missing values, and nominal features were meticulously converted into binary format in order to maximize data representation and enhance model performance.

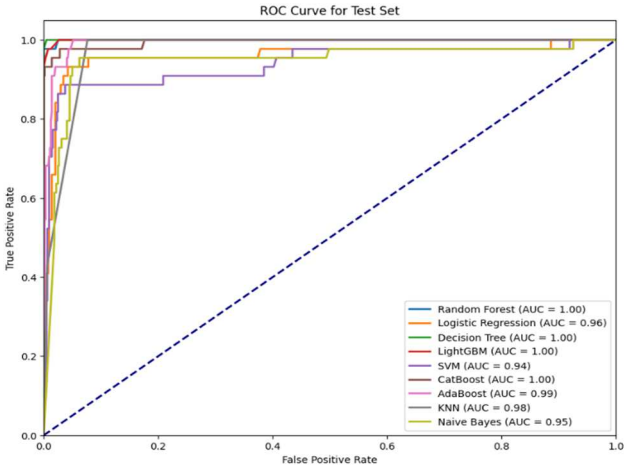


Fig.3. ROC curves of different machine learning classifiers.

This research used the most recent methods put out by K.M. Mohiuddin et al. [8], Tanimu J. et al. [9], Abdoh et al. [10], and Ijaz et al. [32] to assess the prediction ability of our machine learning algorithm for cervical cancer risk. Our strategy combines sophisticated feature selection techniques like SelectKBest, XGBoost, and Chi-Square with ensemble learning to improve model sensitivity, specificity, and accuracy. Methods such as SMOTE-RF, SMOTE-RF RFE, and SMOTE-

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RF-PCA were used by Abdoh et al. [10], whilst Ijaz et al. [11] employed DBSCAN, SMOTETomek, RF, and SMOTE in different configurations. Furthermore, RFE, LASSO, SMOTETomek, and DT techniques were the main topics of Tanimu J. et al. [9].

Yet, Table 3 demonstrates that our strategy outperformed existing approaches in terms of accuracy, attaining an exceptional 99.443% when voting on Random Forest (RF), Decision Tree (DT), and AdaBoost (AdB). Our model also exhibits greater sensitivity than previous approaches.

TABLE-3: COMPARISON TABLE WITH EXISTING WORKS

Author	Dataset	Method	Accuracy
Mohi Uddin et al. [8]	Clinical data (858 patients With 36 features)	RF+MLP	99.19%
Tanimu,J.J [9]	Ref34 (compared against EOC and PAP smear tests)	DT	98.82%
Abdoh et al.[10]	Clinical data (585 patients with 32 features)	SMOTE-RE	96.06%
Ijaz et al. [11]	858 patients with ASCUS	RF	97.72%
Our Proposed Model	Clinical data (585 patients with 36 features)	RF+DT+ AdaBoost	99.44%

V. CONCLUSION

This paper concludes with a description of a new machine learning architecture that greatly improves the precision and dependability of medical diagnostics, especially when estimating the risk of cervical cancer. We have included strong ensemble approaches like Random Forest, AdaBoost, and Decision Tree into our model. A stunning 99.443% accuracy rate is achieved, which is considered to be an outstanding result. The combination of this high accuracy and cutting-edge feature selection methods like SelectKBest, XGBoost, and Chi-Square significantly enhances the precision of diagnosis and customized treatment planning. This AI-driven, scalable approach overcomes issues including feature interpretability and class imbalance, potentially revolutionizing cervical cancer detection. It allows for even greater diagnostic precision and is widely used in a variety of clinical scenarios.

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